Outside Data Access and Agent Web Reading in the AI Law Lab

Which services the lab calls, when its agents go online, and how a site's wishes are honoured

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Abstract

The AI Law Lab runs its language models on university hardware and answers questions from its own versioned libraries, but a research run may also search public legal databases and the open web, and read what it finds. This note sets out every service the lab's server connects to, what is sent to each, and the exact points at which a run goes online. It describes the five legal-database interfaces the lab uses and how agents' calls to them are paced; how every document read is saved into a library before it can be cited; and the checks a web page must pass before a research agent or a crawl may read it. A page is read only if its site permits it: robots.txt is obeyed for the lab's own name and for well-known AI agents, and pages marked "noai" or carrying a text-and-data-mining reservation are left alone. Refusals are reported to the agent and listed on the run's page. The note closes with the safeguards applied to every fetch and the questions that remain open.

1 Introduction

A study of how language models reason about law should be able to say exactly what the model was shown. The lab's runs therefore draw on libraries whose every change is recorded as a numbered version, and each citation in an answer is traced to a document, a library and a version. Reaching outside the libraries threatens both properties: a web page can change or vanish after it is read, and a search sends part of the researcher's question to a third party.

Outside access is nonetheless useful. Agency guidance, recent opinions and scholarship are often not yet in any library. The lab allows it under three conditions. A person must allow it, for each run. Whatever is read is saved into a library first, so that a citation to it names a fixed copy. And a page is read only if its website has not said that it does not want AI systems reading it. Section 2 lists every outside connection; Section 3 describes the legal databases; Section 4 the points at which a research run goes online; Section 5 what people add by hand; Section 6 the checks on a site's wishes; Section 7 the safeguards on every fetch; and Section 9 the limitations.

2 Every outside connection

Table 1 lists every destination the lab's server connects to. Model inference never leaves university hardware, and no prompt, document or answer is sent to a commercial model API. What leaves the server is a search term or an address, never a document a researcher uploaded.

Destination	What is sent	When
The two DGX Spark units (Tailscale, inside the university)	Prompts, and text to embed	Every model call
CourtListener, Federal Register, eCFR, govinfo, SEC EDGAR	A search term, or the identifier of a result	A person searches or adds on the Legal Sources page; an agent searches or reads when legal databases are allowed
Brave Search API	A search query written by an agent	An agent searches the open web, when the open web is allowed
Any public website	A request for one address	An agent reads a page; a page named in the brief is read; a crawl; a person adds or re-checks a link, or drafts a cast from one
UW Shibboleth, InCommon metadata service, Google, Microsoft	Sign-in exchanges, through Keycloak	Someone signs in with that provider

Table 1. Every destination the server connects to, what is sent, and when. The search terms an agent writes are drawn from the research brief, so a brief that must stay confidential should be run with both outside permissions off.

3 The legal databases

3.1 Interfaces and credentials

Five public repositories are reached through the interfaces their publishers provide for programs to use (Table 2). Each is a staging step rather than a live oracle: a search returns candidates, and nothing is relied on until it has been fetched, divided into passages, embedded and saved into a library with its canonical address and the time it was retrieved. Credentials are kept in the server's environment, never in the code or the browser, and a database whose credential is missing reports itself unavailable instead of failing.

Database	Interface and credential	What it holds
CourtListener	REST API v4 [5]; API token, set on the server	Opinions of federal and state courts (Free Law Project). Without a token only the search snippet can be saved, and it is labelled as a snippet.
Federal Register	API v1 [6]; no key	Proposed and final rules, notices and presidential documents since 1994; full text where published, else the abstract, flagged as such.
eCFR	Search and versioner APIs [7]; no key	The Code of Federal Regulations as in force, read as of each title's latest issue.
govinfo (GPO)	govinfo API [8]; api.data.gov key, set on the server	US Code, Public Laws, bills and reports; HTML or text rendition, else the PDF.
SEC EDGAR	Full-text search and the EDGAR archives; no key	Company filings since 2001. Requests name the lab and a contact address, as the SEC's fair-access policy requires [9].

Table 2. The legal databases, the interface used for each, and what it returns.

3.2 Pacing

A person's search on the Legal Sources page goes straight through. Research agents call far more often, several at a time, so each database's requests from agents are spaced at least one second apart across every run on the server. A database that answers HTTP 429 ("too many requests") is left alone for three minutes, and agents are told to search elsewhere meanwhile. One agent search asks each database for four results.

3.3 Open-web search

Brave's Search API [10] answers an agent's web searches, eight results each. Calls are spaced 1.1 seconds apart, the API's per-second limit, and one 429 is retried after two seconds. Results are numbered leads ([W1], [W2], ...): an agent cannot cite a search result, only what it has read.

3.4 Why robots.txt does not govern these

robots.txt addresses crawlers that wander a website. These interfaces exist so that programs can query them; their use is governed by their own terms, keys and rate limits, which the lab follows. The documents they return are largely government works [11] or public filings. That a document is public does not settle every use of it, and researchers remain responsible for theirs.

4 When a research run goes online

4.1 Two permissions

An agentic-workflow run has two permissions, each the researcher's to grant on the plan before the run starts and both off by default: search the legal databases, and search the open web, which includes reading any public page. With both off the run is entirely local. Role-play and document-analysis runs never go online.

4.2 The four stages of a run

With either permission on, network access happens at these points and no others:

- **Planning: none.** The planner drafts sub-questions from the brief on the Spark units. It notes web addresses written in the brief but does not open them.
- **Before research: pages named in the brief.** If the open web is allowed, each address in the brief is read once, before any agent starts, so that every agent has it. These reads pass the same checks as an agent's own (Section 6).
- **During research: the agents' tools.** Each research agent, up to four at once, decides for itself when to call `search_databases`, `search_web` and `read`. Reading is not budgeted; the run's time limit (two hours by default) and the researcher's Stop button end it. A document is read once per run, however many agents ask for it.
- **Write-up: none.** The lead agent writes the answer from the findings and the passages already read.

4.3 How a read becomes a source

Everything read is saved into a library, by default "Fetched: <experiment name>", which records a new version; the run is then pinned to that version, so later library searches by any agent cover the new document. The passages handed back to the agent are numbered in the run's ledger like any other, so a citation to something read online names a document, a library and a version, and stays readable after the page changes. The run's page lists every search with the results it gave, every document read with the library version it was saved to, and every page that was not read because its site does not allow it.

5 What a person adds

On the Legal Sources page a person can search the databases and add chosen results; add web pages by address, up to 20 at a time, and later check them again, which re-reads each page and rebuilds its passages only if the text has changed; or crawl a page's links into a library. In the cast builder a person can draft a role-play from a page given by its address. Adding a link or drafting from one is that person reading the page, as they would in a browser, so it is not put through the checks of Section 6. A crawl, which reads many pages automatically, is.

6 Honouring a site's wishes

Before a research agent reads a page, and before each page of a crawl, the lab asks what the site has published about automated and AI use, and obeys it. Procedure 1 gives the order for an agent; a crawl asks the same questions under its own name.

```
read(address):
  if the address is not public http(s)            -> refuse  (unsafe; nothing fetched)
  rules <- robots.txt and tdmrep.json of the site    (cached up to one hour)
  if robots.txt cannot be read (site failing)     -> refuse
  if robots.txt disallows AILawLab-agent here     -> refuse
  if robots.txt disallows a named AI agent here   -> refuse
  if tdmrep.json reserves TDM rights here         -> refuse
  page <- fetch once as AILawLab-agent/1.0           (at most 8 MB, 45 s)
  if the page moved to another site, check that site's rules too
  if page headers or <meta> say noai,
     or tdm-reservation: 1                        -> refuse  (fetched, not kept)
  save page to the run's library; return its passages, numbered for citation
```

Procedure 1. How a research agent's request to read a page is decided. Each refusal is returned to the agent with its reason, recorded in the trace and listed on the run's page.

6.1 robots.txt, for the lab

The site's robots.txt is read first, following the Robots Exclusion Protocol [1]. Research agents are known to it as `AILawLab-agent` and crawls as `AILawLab-crawler`; a site may write rules for either, or for all crawlers (*). The most specific matching rule decides, and Allow wins a tie. The * and $ wildcards defined by the protocol are understood; Python's built-in robots.txt reader ignores them, which would have let a rule such as `Disallow: /*.pdf$` through. A missing robots.txt sets no rules; one that cannot be read because the site is failing means nothing is read, as the protocol prescribes.

6.2 robots.txt, for AI agents

Many sites now turn named AI agents away while admitting search engines. If a site's robots.txt closes the page to any of the agents in Table 3, the lab treats that as a refusal of AI systems in general, which it is one of. Only rules written for those agents by name count: an agent that falls back to the rules for all crawlers is treated as the lab already is.

Kind	Agents
Assistants reading for a person	ChatGPT-User, Claude-User, Perplexity-User, MistralAI-User
Crawlers of AI companies	GPTBot, OAI-SearchBot, ClaudeBot, Claude-SearchBot, anthropic-ai, PerplexityBot, CCBot, Google-Extended, Applebot-Extended, Bytespider, meta-externalagent, cohere-ai

Table 3. The AI agents whose exclusion from a page the lab honours as its own (`site_rules.AI_AGENTS`). The list is maintained by hand.

6.3 Text-and-data-mining reservations

The TDM Reservation Protocol [2] lets a rights holder reserve text-and-data-mining rights in a machine-readable way, the kind of opt-out European copyright law recognizes [3]. A site may publish the reservation in `/.well-known/tdmrep.json`, by location; a page may carry it in a `tdm-reservation: 1` response header or a `tdm-reservation` meta tag. A reservation means the page is not read, and the licensing address the site gives, if any, is passed on to the agent.

6.4 "noai" on the page

Some sites mark pages "noai", in an `X-Robots-Tag` response header or a robots meta tag in the page's head. The lab honours the directive when it is addressed to every agent or to the lab by name. These markings can be seen only once the page has arrived, so such a page is fetched but neither kept nor shown to the agent.

6.5 When a page is refused

A refused agent is told plainly why and asked to find another source; the refusal is recorded in the run's trace and listed, with its reason, on the run's page. A crawl records the page as "not open to AI use" and carries on with the rest. A site's rules are read at most once an hour per site, across all runs (the protocol allows a day), and a failing site is asked again next time rather than remembered.

7 Safeguards on every fetch

- **Public addresses only.** The server can reach the Spark units, its database and the university network, so a link is fetched only if it uses `http` or `https` and its host resolves exclusively to public addresses. Every redirect, up to five, is checked again.
- **Bounded.** A response is cut off at 8 MB and 45 seconds. A page kept in a library keeps up to 100,000 words; a source for a cast, 12,000.
- **Named.** Requests say who is asking: `AILawLab-agent/1.0` for agents and `AILawLab-crawler/1.0` for crawls, each with the lab's address and a contact, so that a site owner can see who is reading and how to reach the lab.
- **Polite crawling.** A crawl stays on one website and goes one level deep. It fetches one page at a time at least two seconds apart, longer if robots.txt asks (up to 30 seconds), takes at most 100 pages, stops on HTTP 429 or 503 or after three failures in a row, and

runs one at a time per site. It stores the licence or copyright notice it finds with every document.

- **Recorded.** Searches with their results, documents read with their library versions, pages refused with their reasons, and every crawl with the rules it ran under and the outcome for each link are kept in the lab's database.

8 The lab's own interface

The browser talks to the lab through JSON endpoints under `/ai_law_lab/api/`. They are the portal's plumbing, not a public API: every one needs a signed-in, approved account, and changes need the researcher or administrator role. The endpoints that reach outside are those for database search and ingest (`/api/sources/search`, `/api/sources/ingest`), web links (`/api/sources/links` and the two refresh endpoints), crawls (`/api/crawls`, `/api/crawls/preview`) and cast sources (`/api/source/read`).

9 Limitations and open questions

- **The AI-agent list is maintained by hand.** New agents appear regularly; the list in Table 3 should be reviewed each term.
- **Machine-readable AI preferences are still being standardized.** The IETF AI Preferences working group [4] is defining a common vocabulary. The lab follows robots.txt, "noai" and TDMRep, which sites use today, and should adopt the IETF vocabulary once it settles.
- **Permission to read is not a licence to reuse.** The licence notice a crawl records, and the address and retrieval time kept with every document, let a researcher check before relying on, quoting or publishing what was collected.
- **Pages a person adds by hand are not checked** (Section 5). If the lab's practice should be stricter, the same check can be applied there.

Appendix: where it lives

File	What it does
`sources.py`	The five legal databases: search and fetch
`network_tools.py, tools.py`	Agents' searching and reading; pacing; the fetch library
`research.py, experiments.py`	Run permissions; pages named in the brief
`site_rules.py`	robots.txt, AI agents, TDMRep and noai checks
`source_material.py`	Fetching, the public-address guard, size and time limits
`web_links.py`	Web pages kept in a library, and checking them again
`crawler.py`	Crawling a page's links into a library
`web/views.py, web/templates/run.html`	The run page: read online, and refused

Table 4. Source files, under `src/ailawlab/`. Settings quoted in this note are defaults from `config.py`, each overridable by an `AILAWLAB_` environment variable. Tests: `tests/test_site_rules.py`, `tests/test_network_tools.py`, `tests/test_crawler.py`.

References

[1] M. Koster, G. Illyes, H. Zeller and L. Sassman. Robots Exclusion Protocol. RFC 9309, Internet Engineering Task Force, September 2022. https://www.rfc-editor.org/rfc/rfc9309

[2] W3C Text and Data Mining Reservation Protocol Community Group. TDM Reservation Protocol (TDMRep). Final Community Group Report. https://www.w3.org/community/tdmrep/

[3] Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on copyright and related rights in the Digital Single Market, Article 4.

[4] IETF AI Preferences (aipref) Working Group. https://datatracker.ietf.org/wg/aipref/about/

[5] Free Law Project. CourtListener REST API, v4. https://www.courtlistener.com/help/api/rest/

[6] Office of the Federal Register. Federal Register API, v1. https://www.federalregister.gov/developers/documentation/api/v1

[7] Office of the Federal Register. eCFR API. https://www.ecfr.gov/developers/documentation/api/v1

[8] US Government Publishing Office. govinfo API. https://api.govinfo.gov/docs/

[9] US Securities and Exchange Commission. Accessing EDGAR Data. https://www.sec.gov/os/accessing-edgar-data

[10] Brave Software. Brave Search API. https://api.search.brave.com/app/documentation

[11] 17 U.S.C. § 105, Subject matter of copyright: United States Government works.